

with rapid detection capabilities, swiftly identifying and addressing security threats within the CSP environments.

However, it's essential to weigh these benefits against the limitations. While cloud-native tools cover basic security needs well, they lack the ability to be customised based on an organisation's needs. This lack of customisation can be a significant drawback, particularly for businesses operating in highly regulated industries like banks and federal services. Additionally, relying completely on CSP-provided controls may lead to vendor lock-in, making it challenging to migrate to other cloud providers or third-party commercial solutions in the future, especially for organisations considering a multi-cloud strategy.

In recent times, CSPs have recognised these limitations and have begun offering cloud-native solutions that support a multi-cloud setup as well. A prime example of such solutions is Azure's Defender for Cloud. Ultimately, businesses that embrace these solutions are likely to become more profitable in the long run.

Third-party commercial tools

On the other hand, third-party commercial solutions offer a range of features and functionalities. They often offer superior threat detection capabilities, enhanced compliance management tools, and greater customisation options. These solutions can seamlessly integrate with existing security systems and are compatible with multi-cloud environments, providing consistent security measures across various providers. They also provide deeper insights beyond what CSPs typically offer.

However, there are notable challenges associated with third-party commercial solutions; they bring their own risks into the supply chain. Integrating and managing these solutions can be complex, requiring additional training and resources.

Moreover, the costs associated with licensing and maintaining third-party commercial solutions can be substantial, particularly for organisations with large-scale cloud deployments. These expenses add to the overall opex budget allocated for cloud security.

Open source tools

Open source tools also play a significant role in cloud security. These tools, developed collaboratively by communities of developers, offer flexibility, cost-effectiveness, and support. They provide access to a range of security functionalities without the huge price tag associated with cloud-native and third-party tools.

However, leveraging open source tools may require technical expertise for integration, customisation and management. Organisations must ensure proper community support and maintenance of the tool to maximise the benefits. Industries with strict regulations like banks and federal services often refrain from utilising open source solutions due to their concerns regarding security governance. Security is the biggest limitation in open source tools.

This leads to the crucial question: Why do organisations choose open source tools, and why do consultants recommend them? The answer is straightforward: it's the calculation of benefits over risks taken. Organisations are aware of the advantages of utilising open source tools over the potential risks involved. This is the same thought process behind the shift to cloud computing. Despite the known security challenges associated in migrating to the cloud, many organisations have proceeded with the migration, driven by the assessment that the benefits of cloud adoption surpass the associated risks.

As a consultant, before recommending cloud security tools I thoroughly try to understand six key areas.

Business requirements: A project fails if it doesn't align to the organisation's business requirements and goals. This includes understanding the organisation's industry, regulatory requirements, and specific security concerns. The time invested in understanding the business requirements during client engagements helps to make specific recommendations that align closely with the organisation's goals and objectives. However, this level of detail can also prolong the engagement process.

Team's knowledge and experience: Next, it is time to assess the team's ability to handle cloud security tools. Financial institutions, for example, prefer straightforward third-party tools because they streamline processes for risk and compliance teams. In contrast, startups with proficient teams may lean towards using open source tools for their flexibility and cost options.

Cost and budget considerations: Understanding the cost and budget considerations ensures that tool recommendations align with both technical needs and financial limits. This way, we find a good balance between how well the security works and how much it costs, leading to smart, long-lasting choices for using cloud security tools.

Tool features and integration: Matching the chosen tool features with the organisation's needs is critical. Companies operating on a single cloud platform might benefit more from using native tools provided by their cloud service provider. However, organisations with multi-cloud setups often require third-party tools for better management. Additionally, integration with other systems is key, and third-party tools typically have an advantage in this area.

Preventive and protection controls: The priority is to select tools that can both prevent and respond to security threats effectively. This often involves combining native tools with third-party